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Summary

Final-year **Computer Science & Engineering** student with hands-on experience in **Full-Stack Development, Generative AI, Machine Learning, Data Engineering, Scalable Backend Systems, and Cloud-Native Applications**. Experienced in building **AI-powered applications** using **LLMs, RAG, Prompt Engineering, Embeddings, Vector Search, and AI Agents**, alongside production-grade web applications and RESTful APIs using **Python, C++, Java, SQL, JavaScript/TypeScript, React.js, Next.js, Node.js, and Express.js**. Proficient in **PostgreSQL, Docker, AWS, GCP, Linux, and CI/CD**, with a strong foundation in **Data Structures & Algorithms, DBMS, Distributed Systems, Software Engineering, and DevOps**. Holds industry-recognized certifications in **Cloud Computing, Machine Learning, SQL, Data Engineering, GitHub, and Python Data Analytics**.

Education

Kalinga Institute of Industrial Technology <i>Bachelor of Technology in Computer Science & Engineering</i>	July 2023 - June 2027 Bhubaneswar, Odisha
• Relevant Coursework: Introduction to C Programming, Data Structures and Algorithms (C), Object Oriented Programming (Java), Operating Systems, Database Management System (Oracle Database Express Edition XE), Computer Network, Design and Analysis of Algorithms, Software Engineering • Relevant Elective: Linux System Administration (Red Hat Enterprise Linux), Computational Intelligence, High Performance Computing (MIPS), Cloud Computing, Deep Learning	
Royal Public Sr. Sec. School <i>Intermediate in Non-Medical with Computer Science</i>	April 2021 - April 2022 Gurgaon, Haryana
• Relevant Coursework: Introduction to Python Programming, Implementing Basic Data Structures in Python, Database Management System (MySQL), Connecting MySQL to Python	

Experience

Technical Societies, Kalinga Institute of Industrial Technology <i>Full-Stack Developer & DevOps Engineer (Google Developer Group, GeeksforGeeks & IoT Lab)</i>	Feb 2025 – Present Bhubaneswar, Odisha
• Developed and deployed 5+ production-grade full-stack applications using React.js, Next.js, Node.js, Express.js, Docker, AWS, and PostgreSQL, serving 2,000+ users. • Designed and implemented secure RESTful APIs, Role-Based Access Control (RBAC), and cloud-native deployment pipelines while collaborating with a team of 8+ developers. • Mentored 10+ students in Full-Stack Development, DevOps, Docker, and AWS, while contributing to hackathons, workshops, and university-wide technical initiatives.	

Projects

GradeX LMS – AI-Powered Learning Management System <i>Next.js, TypeScript, Node.js, Drizzle ORM, PostgreSQL, AWS</i>	
• Designed and implemented a multi-tenant Learning Management System with secure Role-Based Access Control (RBAC) supporting 4 user roles, authentication, and session management. • Built an AI-powered assessment engine using the Gemini API for automated quiz generation and assignment evaluation. • Developed scalable cloud-native services integrating AWS SES, PostgreSQL, and Cloudflare R2 for automated notifications and secure file storage.	
Course-Aware AI Learning Platform <i>Python, FastAPI, LangChain, LangGraph, Gemini, RAG, pgvector, PostgreSQL, Next.js</i>	
• Engineered a course-aware RAG system that ingests lecture slides, notes, PDFs, and syllabi to provide source-grounded answers using semantic retrieval, embeddings, metadata filtering, and pgvector. • Developed an agentic study assistant using LangGraph and Gemini with specialized workflows for QA, flashcard generation, quiz creation, and syllabus-aware exam preparation. • Implemented structured LLM outputs, prompt templates, conversation memory, source attribution, and hallucination safeguards to generate reliable, course-specific responses aligned with uploaded learning materials.	
University Schedule Orchestration Platform <i>Next.js, Node.js, TypeScript, Express.js, PostgreSQL, Drizzle ORM, Docker, AWS</i>	
• Built a production-grade scheduling platform serving 2,000+ students with scalable RESTful APIs using Node.js, Express.js, and PostgreSQL.	

- Containerized and deployed the application on **AWS EC2** using **Docker** and **Nginx**, achieving **100% uptime** with automated health checks.

PrepCrate – AI Academic Resource Platform | *Next.js, TypeScript, Node.js, PostgreSQL, Drizzle ORM, Cloudflare R2, Gemini API*

- Developed a centralized academic resource platform enabling students to securely upload, organize, and access **500+ study materials**.
- Integrated the **Gemini API** to automatically extract and structure key metadata from uploaded materials, including titles, subjects, topics, and descriptions, and persist the generated information in **PostgreSQL**.
- Implemented a responsive **Next.js** frontend with **Cloudflare R2** for scalable document storage and **Drizzle ORM** for type-safe database operations.

Single Image Super-Resolution using CNN & GAN | *Python, PyTorch, OpenCV*

- Developed a deep learning pipeline combining **CNN-based image reconstruction** and **GAN-based refinement** to recover high-resolution images from low-resolution inputs while preserving fine-grained visual details.
- Designed custom **adversarial and perceptual loss functions** to balance pixel-level accuracy and perceptual quality, improving texture preservation and visual sharpness during model optimization.

Distributed File Storage System | *Node.js, TypeScript, Express.js, PostgreSQL, Redis, Docker*

- Built a **fault-tolerant distributed storage system** with chunk replication, metadata indexing, **consistent hashing**, and automated replica synchronization across multiple nodes.
- Implemented **heartbeat monitoring**, SHA-256 integrity validation, and Redis caching; containerized services with **Docker Compose**, reducing metadata lookup latency by approximately **60%**.

Technical Skills

- **Programming Languages:** Python, C++, Java, C, SQL, TypeScript, JavaScript, HTML5, CSS3, SCSS
- **Frameworks & Libraries:** React.js, Next.js, Node.js, Express.js, FastAPI, LangChain, LangGraph, Tailwind CSS, Scikit-learn, TensorFlow, Keras, PyTorch, Pandas, NumPy
- **Databases & ORMs:** PostgreSQL, MySQL, MongoDB, Oracle SQL, Firebase Firestore, Supabase, Neon, pgvector, Drizzle ORM, Prisma ORM, Mongoose
- **Cloud, DevOps & Infrastructure:** AWS (EC2, S3, SES), Google Cloud Platform (Cloud Run, App Engine, Compute Engine, Cloud Storage, Cloud Functions), Oracle Cloud Infrastructure (OCI), Docker, Docker Compose, Nginx, Cloudflare R2, GitHub Actions, CI/CD, Linux, Bash/Shell Scripting, Virtual Private Network (VPC)
- **Authentication & Security:** OAuth 2.0, JWT, Auth.js, Clerk, Firebase Authentication, Role-Based Access Control (RBAC)
- **AI, Generative AI & Machine Learning:** Open API, Gemini API, Large Language Models (LLMs), Prompt Engineering, Embeddings, Vector Search, Semantic Search, Retrieval-Augmented Generation (RAG), AI Agents, Agentic AI, Tool Calling, Structured Outputs, Scikit-learn, TensorFlow, PyTorch, Artificial Neural Networks, Deep Learning, Natural Language Processing (NLP)
- **Developer Tools & Platforms:** Git, GitHub, GitHub Actions, Postman, VS Code, Neovim, Arduino IDE, ESP32, STM32
- **Software Engineering Concepts:** Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, Operating Systems, Computer Networks, Distributed Systems, RESTful APIs, API Development, Microservices, System Design, MVC Architecture, Reverse Proxy, Load Balancing, Concurrency, Software Development Life Cycle (SDLC)

Certifications

- **Cloud Computing**
 - Google Cloud Engineer Professional Certificate *Google Cloud – Coursera*
 - Developing Applications with Google Cloud *Google Cloud – Coursera*
 - Architecting with Google Compute Engine *Google Cloud – Coursera*
 - OCI Foundations Associate *Oracle Cloud Infrastructure*
 - OCI AI Foundations Associate *Oracle Cloud Infrastructure*
- **Data Engineering & Analytics**
 - Data Engineer Associate *DataCamp*
 - SQL Associate *DataCamp*
 - Python Data Associate *DataCamp*
 - Google Data Analytics with Python *Google – Coursera*
- **Software Engineering**
 - GitHub Foundations *GitHub*
 - Red Hat System Administration I (RH124) *Red Hat*
- **Artificial Intelligence**
 - Introduction to Generative AI Learning Path Specialization *Google – Coursera*
 - Machine Learning Foundations: A Case Study Approach *University of Washington – Coursera*
 - Machine Learning Specialization *DeepLearning.AI & Stanford Online – Coursera*